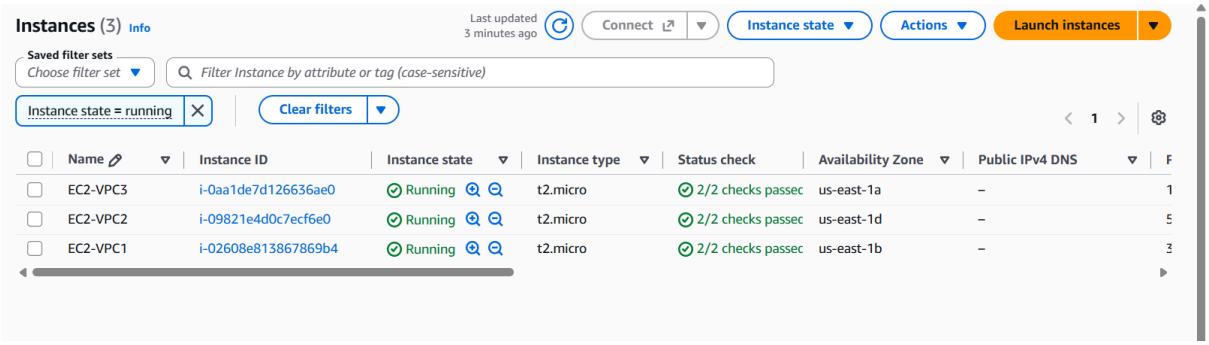
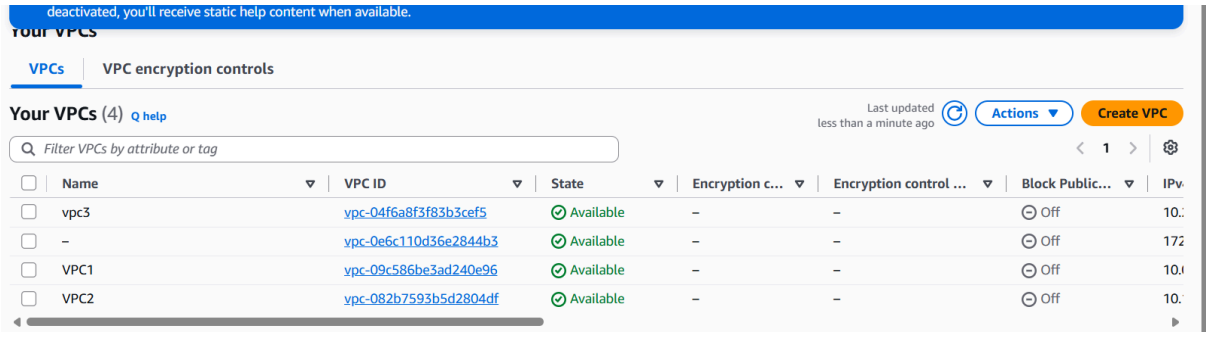
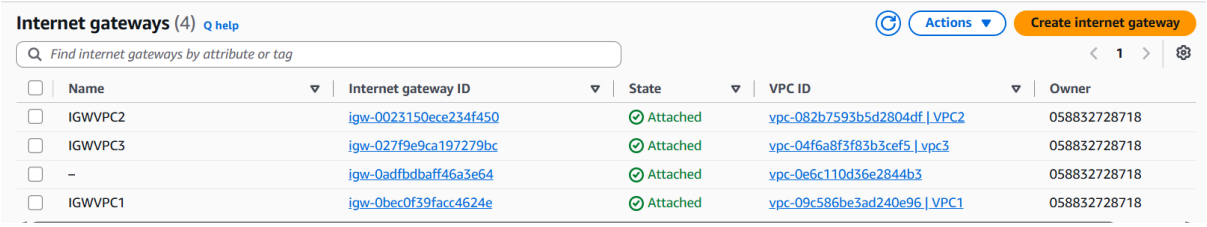
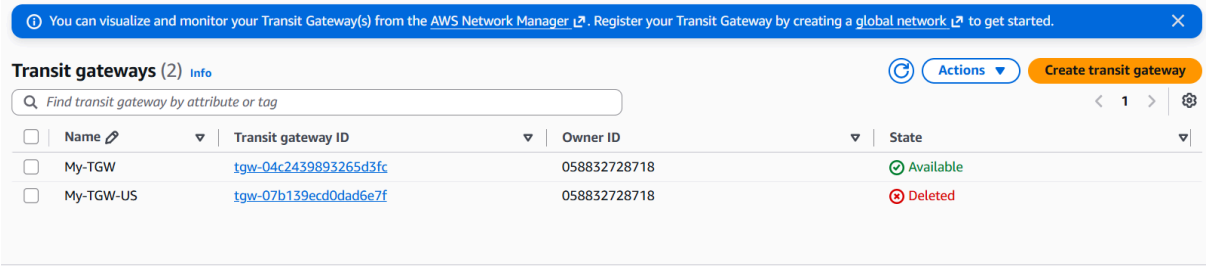
Objectives:

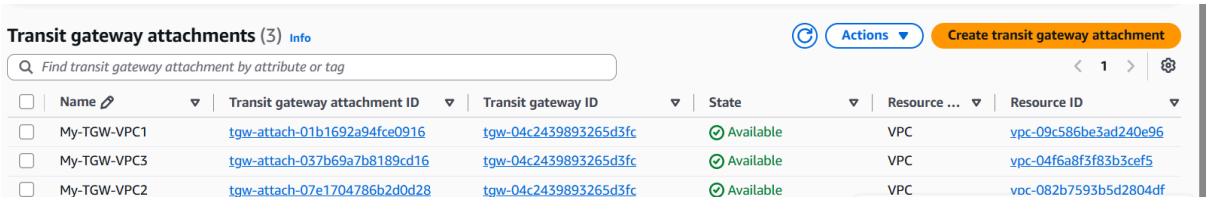
- **Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.**











ec2-user@ip-10-0-0-193:~

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~ (master)
$ cd ~/Downloads

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/Downloads (master)
$ ssh -i VPC1.pem ec2-user@34.239.104.43
gnssh: connect to host 34.239.104.43 port 22: Connection timed out

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/Downloads (master)
$ ssh -i VPC1.pem ec2-user@34.239.104.43
ssh: connect to host 34.239.104.43 port 22: Connection timed out

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/Downloads (master)
$ ssh -i VPC1.pem ec2-user@34.239.104.43
ssh: connect to host 34.239.104.43 port 22: Connection timed out

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/Downloads (master)
$ ssh -i VPC1.pem ec2-user@34.239.104.43
The authenticity of host '34.239.104.43 (34.239.104.43)' can't be established.
ED25519 key fingerprint is SHA256:Yz/oPNivfkJxjKjaOJZQtLeIyDQktHFH2bTo21HDffY.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.239.104.43' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-0-193 ~]$ ping 10.1.1.193
PING 10.1.1.193 (10.1.1.193) 56(84) bytes of data.
^[[A^[[A^C
--- 10.1.1.193 ping statistics ---
160 packets transmitted, 0 received, 100% packet loss, time 165375ms
```

```
[ec2-user@ip-10-0-0-193 ~]$ ping 10.1.1.193
PING 10.1.1.193 (10.1.1.193) 56(84) bytes of data.
^[[A^[[A^C
--- 10.1.1.193 ping statistics ---
160 packets transmitted, 0 received, 100% packet loss, time 165375ms

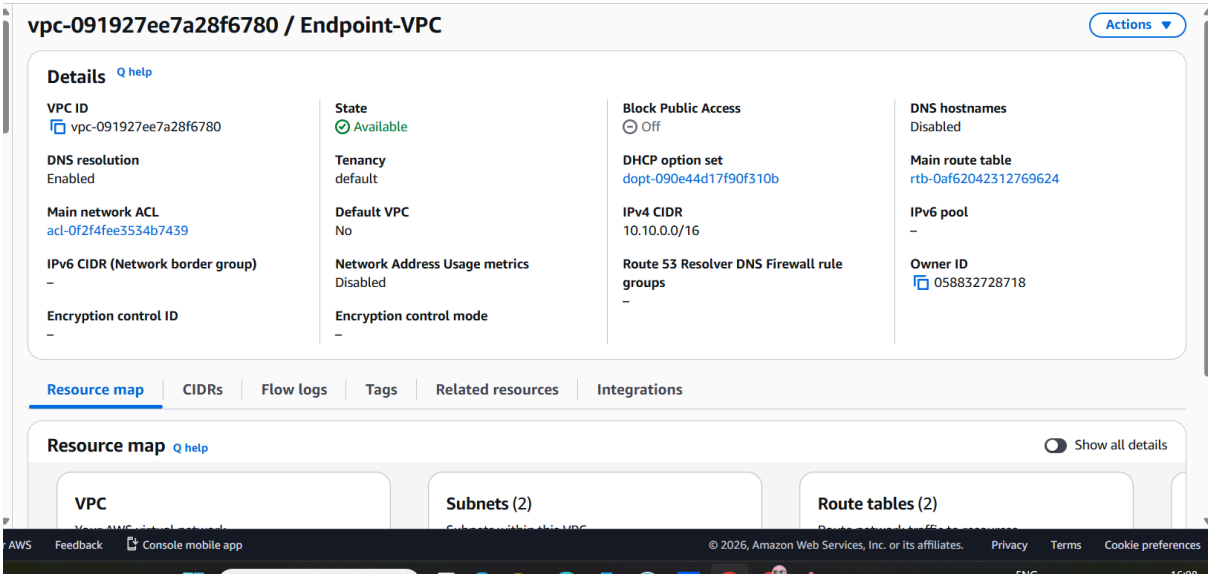
[ec2-user@ip-10-0-0-193 ~]$ ping 10.1.1.193
PING 10.1.1.193 (10.1.1.193) 56(84) bytes of data.
64 bytes from 10.1.1.193: icmp_seq=43 ttl=126 time=2.42 ms
64 bytes from 10.1.1.193: icmp_seq=44 ttl=126 time=1.31 ms
64 bytes from 10.1.1.193: icmp_seq=45 ttl=126 time=1.30 ms
64 bytes from 10.1.1.193: icmp_seq=46 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=47 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=48 ttl=126 time=1.31 ms
64 bytes from 10.1.1.193: icmp_seq=49 ttl=126 time=1.34 ms
64 bytes from 10.1.1.193: icmp_seq=50 ttl=126 time=1.33 ms
64 bytes from 10.1.1.193: icmp_seq=51 ttl=126 time=1.37 ms
64 bytes from 10.1.1.193: icmp_seq=52 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=53 ttl=126 time=1.94 ms
64 bytes from 10.1.1.193: icmp_seq=54 ttl=126 time=1.31 ms
64 bytes from 10.1.1.193: icmp_seq=55 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=56 ttl=126 time=1.31 ms
64 bytes from 10.1.1.193: icmp_seq=57 ttl=126 time=1.69 ms
64 bytes from 10.1.1.193: icmp_seq=58 ttl=126 time=1.33 ms
64 bytes from 10.1.1.193: icmp_seq=59 ttl=126 time=1.35 ms
64 bytes from 10.1.1.193: icmp_seq=60 ttl=126 time=1.31 ms
64 bytes from 10.1.1.193: icmp_seq=61 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=62 ttl=126 time=1.30 ms
64 bytes from 10.1.1.193: icmp_seq=63 ttl=126 time=1.74 ms
64 bytes from 10.1.1.193: icmp_seq=64 ttl=126 time=1.45 ms
64 bytes from 10.1.1.193: icmp_seq=65 ttl=126 time=1.34 ms
64 bytes from 10.1.1.193: icmp_seq=66 ttl=126 time=1.29 ms
64 bytes from 10.1.1.193: icmp_seq=67 ttl=126 time=1.29 ms
64 bytes from 10.1.1.193: icmp_seq=68 ttl=126 time=1.30 ms
64 bytes from 10.1.1.193: icmp_seq=69 ttl=126 time=1.31 ms
64 bytes from 10.1.1.193: icmp_seq=70 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=71 ttl=126 time=1.31 ms
64 bytes from 10.1.1.193: icmp_seq=72 ttl=126 time=1.36 ms
64 bytes from 10.1.1.193: icmp_seq=73 ttl=126 time=1.59 ms
64 bytes from 10.1.1.193: icmp_seq=74 ttl=126 time=1.54 ms
64 bytes from 10.1.1.193: icmp_seq=75 ttl=126 time=1.29 ms
64 bytes from 10.1.1.193: icmp_seq=76 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=77 ttl=126 time=1.43 ms
64 bytes from 10.1.1.193: icmp_seq=78 ttl=126 time=1.88 ms
64 bytes from 10.1.1.193: icmp_seq=79 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=80 ttl=126 time=1.34 ms
```

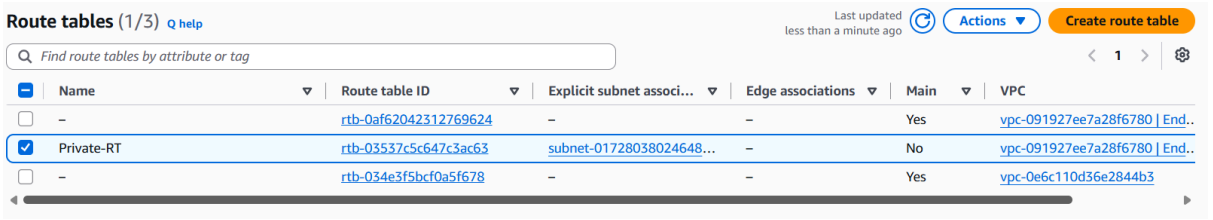
```
64 bytes from 10.1.1.193: icmp_seq=74 ttl=126 time=1.54 ms
64 bytes from 10.1.1.193: icmp_seq=75 ttl=126 time=1.29 ms
64 bytes from 10.1.1.193: icmp_seq=76 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=77 ttl=126 time=1.43 ms
64 bytes from 10.1.1.193: icmp_seq=78 ttl=126 time=1.88 ms
64 bytes from 10.1.1.193: icmp_seq=79 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=80 ttl=126 time=1.34 ms
64 bytes from 10.1.1.193: icmp_seq=81 ttl=126 time=1.32 ms
64 bytes from 10.1.1.193: icmp_seq=82 ttl=126 time=1.66 ms
64 bytes from 10.1.1.193: icmp_seq=83 ttl=126 time=1.61 ms
64 bytes from 10.1.1.193: icmp_seq=84 ttl=126 time=1.47 ms
64 bytes from 10.1.1.193: icmp_seq=85 ttl=126 time=1.30 ms
^C
--- 10.1.1.193 ping statistics ---
85 packets transmitted, 43 received, 49.4118% packet loss, time 85763ms
rtt min/avg/max/mdev = 1.288/1.425/2.415/0.221 ms
[ec2-user@ip-10-0-0-193 ~]$ ping 10.2.1.153
PING 10.2.1.153 (10.2.1.153) 56(84) bytes of data.
64 bytes from 10.2.1.153: icmp_seq=1 ttl=126 time=2.55 ms
64 bytes from 10.2.1.153: icmp_seq=2 ttl=126 time=0.876 ms
64 bytes from 10.2.1.153: icmp_seq=3 ttl=126 time=0.922 ms
64 bytes from 10.2.1.153: icmp_seq=4 ttl=126 time=0.884 ms
64 bytes from 10.2.1.153: icmp_seq=5 ttl=126 time=0.862 ms
64 bytes from 10.2.1.153: icmp_seq=6 ttl=126 time=0.849 ms
64 bytes from 10.2.1.153: icmp_seq=7 ttl=126 time=1.11 ms
```

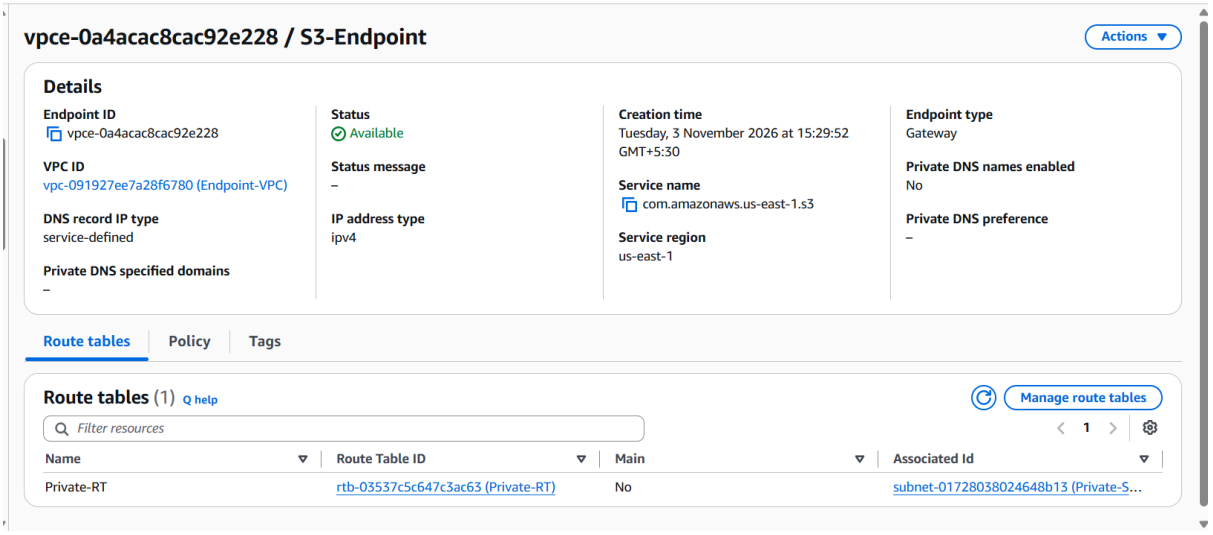
1. I created **3 VPCs**:

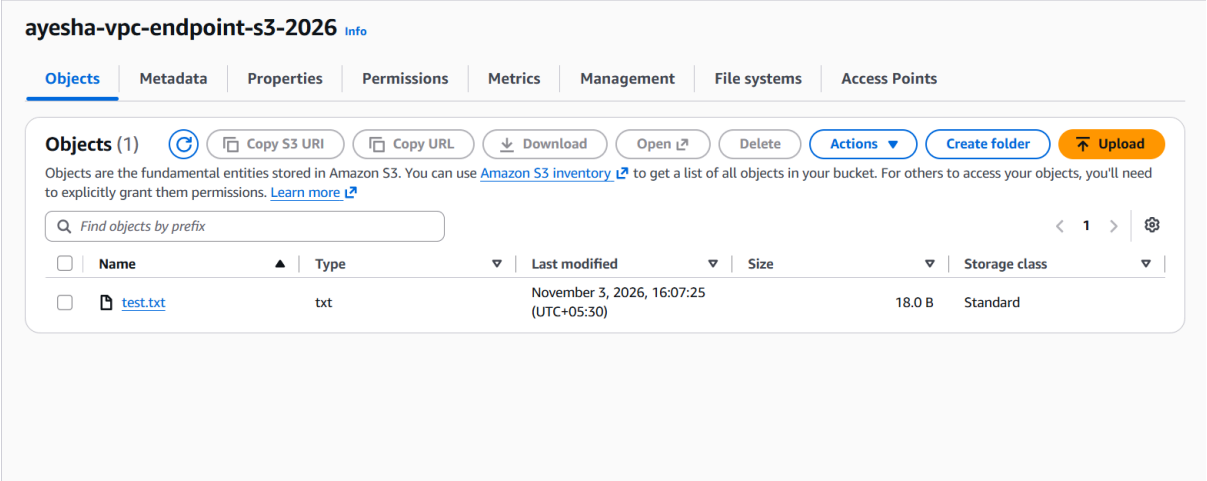
    - VPC1: `10.0.0.0/24`
    - VPC2: `10.1.0.0/16`
    - VPC3: `10.2.0.0/16`
2. I created **subnets** in each VPC.
3. I launched **one EC2 instance in each VPC**.
4. I created a **Transit Gateway**.
5. I waited until the Transit Gateway became **Available**.
6. I created **3 VPC Transit Gateway attachments**:
    - VPC1 → TGW
    - VPC2 → TGW
    - VPC3 → TGW
7. I waited until all three attachments became **Available**.
8. I verified the **Transit Gateway route table**.
9. I created route tables:
    - `VPC1-TGW-RT`
    - `VPC2-TGW-RT`
    - `VPC3-TGW-RT`
10. I associated each route table with its respective subnet.
11. I added VPC2 and VPC3 routes to the **VPC1 route table**, targeting the Transit Gateway.
12. I added VPC1 and VPC3 routes to the **VPC2 route table**, targeting the Transit Gateway.
13. I added VPC1 and VPC2 routes to the **VPC3 route table**, targeting the Transit Gateway.
14. I configured the **Security Groups** to allow SSH and ICMP traffic.
15. I connected to the EC2 instance using **SSH**.
16. I used the **private IP addresses** of the other EC2 instances to test connectivity.
17. I verified connectivity using **ping** between the three VPCs.
18. I captured screenshots of the **VPCs, TGW, attachments, route tables, and successful ping results** as evidence.

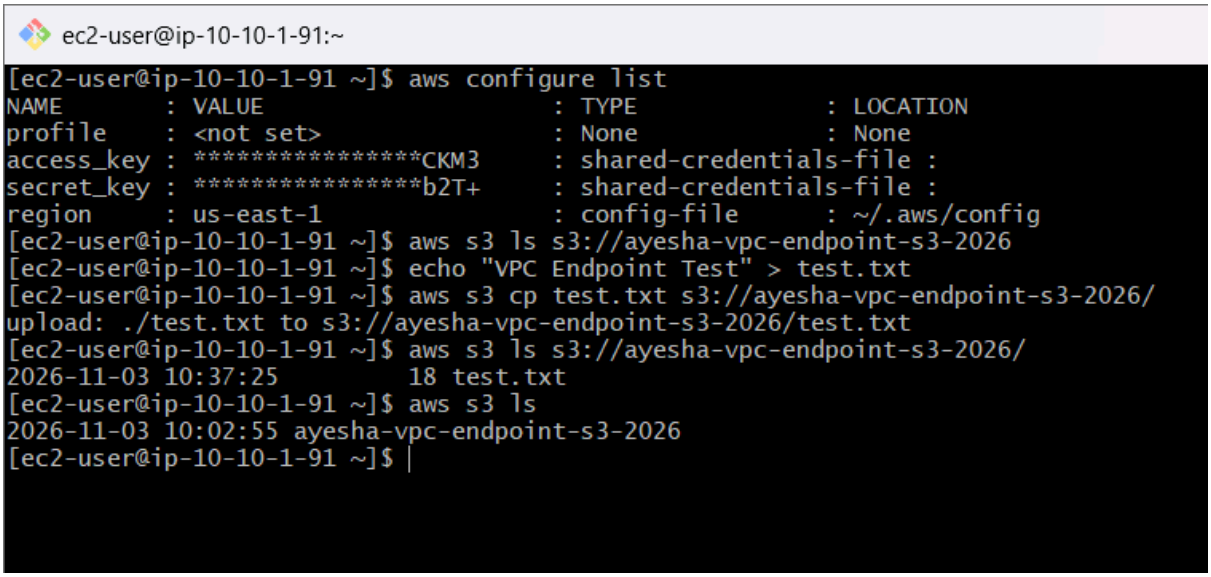
2. Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats

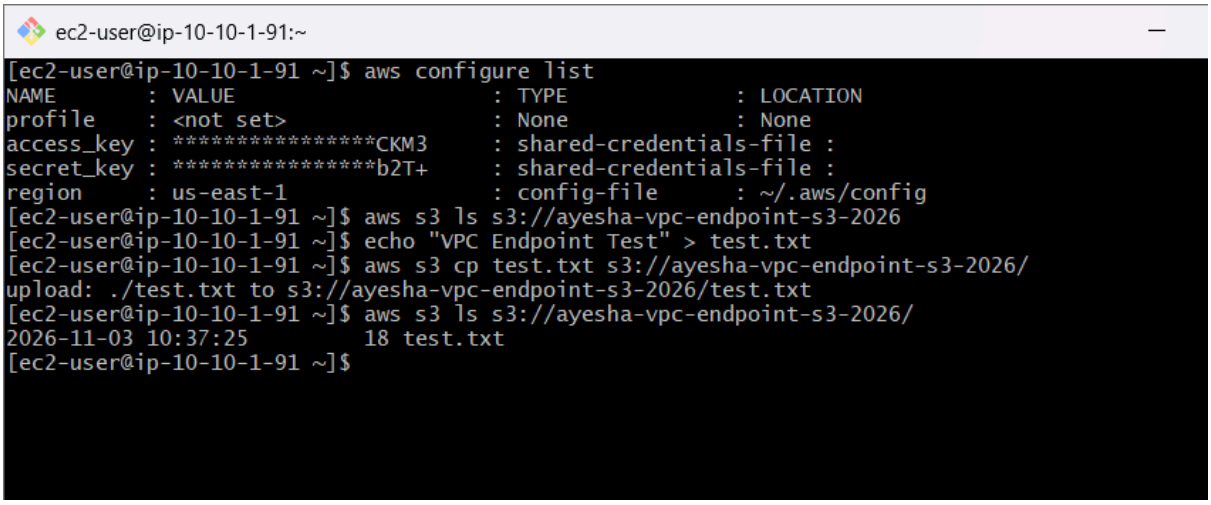

vpc-091927ee7a28f6780 / Endpoint-VPC
Actions
Details
Q help
VPC ID
vpc-091927ee7a28f6780
State
Available
Block Public Access
Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-090e44d17f90f310b
Main route table
rtb-0af62042312769624
Main network ACL
acl-0f2f4fee3534b7439
Default VPC
No
IPv4 CIDR
10.10.0.0/16
IPv6 pool
–
IPv6 CIDR (Network border group)
–
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
–
Owner ID
058832728718
Encryption control ID
–
Encryption control mode
–
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Resource map
Q help
Show all details
VPC
Subnets (2)
Route tables (2)
AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Route tables (1/3)
Q help
Last updated
less than a minute ago
Actions
Create route table
Find route tables by attribute or tag
1
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
–
rtb-0af62042312769624
–
–
Yes
vpc-091927ee7a28f6780 | End..
Private-RT
rtb-03537c5c647c3ac63
subnet-01728038024648...
–
No
vpc-091927ee7a28f6780 | End..
–
rtb-034e3f5bcf0a5f678
–
–
Yes
vpc-0e6c110d36e2844b3


vpce-0a4acac8cac92e228 / S3-Endpoint
Actions
Details
Endpoint ID
vpce-0a4acac8cac92e228
Status
Available
Creation time
Tuesday, 3 November 2026 at 15:29:52 GMT+5:30
Endpoint type
Gateway
VPC ID
vpc-091927ee7a28f6780 (Endpoint-VPC)
Status message
–
Service name
com.amazonaws.us-east-1.s3
Private DNS names enabled
No
DNS record IP type
service-defined
IP address type
ipv4
Service region
us-east-1
Private DNS preference
–
Private DNS specified domains
–
Route tables
Policy
Tags
Route tables (1)
Q help
Manage route tables
Filter resources
1
Name
Route Table ID
Main
Associated Id
Private-RT
rtb-03537c5c647c3ac63 (Private-RT)
No
subnet-01728038024648b13 (Private-S...







1. I created the Endpoint-VPC with CIDR `10.10.0.0/16`.
2. I created a public subnet `10.10.0.0/24`.
3. I created a private subnet `10.10.1.0/24`.
4. I created an Internet Gateway and attached it to `Endpoint-VPC`.

5. I created a public route table and added:
   `0.0.0.0/0 → Internet Gateway`.
6. I associated the public subnet with the public route table.
7. I created a private route table and associated it with the private subnet.
8. I launched a Public EC2 in the public subnet with a public IPv4 address.
9. I launched a Private EC2 in the private subnet with no public IPv4 address.
10. I configured the Private EC2 security group to allow SSH 22 from the Public EC2 security group.
11. I connected from Git Bash → Public EC2 → Private EC2.
12. I created the S3 bucket:
    `ayesha-vpc-endpoint-s3-2026`
13. I created an S3 Gateway VPC Endpoint.
14. I selected:
    - VPC: `Endpoint-VPC`
    - Service: S3
    - Type: Gateway
    - Route table: Private route table
15. I created the endpoint and waited until it became Available.
16. I checked the private route table and verified that the S3 prefix-list route points to the VPC endpoint.
17. I configured the required S3 IAM permissions for my test IAM user.
18. From the Private EC2, I tested:

```
None
aws s3 ls s3://ayesha-vpc-endpoint-s3-2026
```

19. I created a test file:

```
None
echo "VPC Endpoint Test" > test.txt
```

20. I uploaded the file:

```
None
aws s3 cp test.txt s3://ayesha-vpc-endpoint-s3-2026/
```

21. I verified the file:

```
None
aws s3 ls s3://ayesha-vpc-endpoint-s3-2026/
```

22. **I captured screenshots of the VPC, subnets, EC2s, S3 bucket, endpoint, route table, and successful S3 test as evidence.**